



Title

lpbwcode — Bandwidth selection for local polynomial conditional density estimation.

Syntax

```
lpbwcode yvar xvars [if] [in] [, grid(var) xeval(numlist) p(#) q(#) mu(#) nu(#)
kernel(KernelFn) bwselect(BwMethod) ng(#) gridspacing(Spacing) noregularize
genvars(prefix) rgrid(var) rindex(var) separator(#) ]
```

Description

lpbwcode implements rule-of-thumb bandwidth selection for local polynomial conditional density estimation. The first variable is the outcome and all remaining variables are conditioning covariates. Numerical calculations are implemented in standalone Mata code.

Companion command: **lpcde** for estimation and inference.

Options

grid(*var*) specifies the y-grid for bandwidth selection. When omitted, the R package default grid is used.

xeval(*numlist*) specifies the conditioning value. The list must contain one number for each conditioning covariate. When omitted, componentwise medians are used.

p(#) and **q**(#) specify local polynomial orders in the y- and x-directions. Defaults are **p**(2) and **q**(1).

mu(#) and **nu**(#) specify derivative orders with respect to y and x. Defaults are **mu**(1) and **nu**(0) subject to the polynomial orders.

kernel(*KernelFn*) specifies **epanechnikov**, **triangular**, or **uniform**. Default is **epanechnikov**.

bwselect(*BwMethod*) specifies **imse-rot** or **mse-rot**. Default is **imse-rot**.

ng(#) and **gridspacing**(*Spacing*) control automatic grid generation when **grid**() is omitted. **gridspacing**(**quantile**) requests quantile spacing.

noregularize suppresses the local sample size regularization used by the default bandwidth selector.

genvars(*prefix*) stores double-precision results in variables named *prefix_grid*, *prefix_bw*, and *prefix_nh*.

rgrid(*var*) displays rows nearest to the supplied y-grid. **rindex**(*var*) displays rows nearest to supplied row indices.

separator(#) draws a separator line after every # rows in the display.

Saved Results

lpbwcode saves the following in **e()**:

e(result)	matrix with columns y_grid bw nh index
e(N)	sample size
e(d)	number of conditioning covariates
e(p) , e(q) , e(mu) , e(nu)	polynomial and derivative orders

Examples

```
. set obs 500
. set seed 42
. gen double x = rnormal()
. gen double y = x + rnormal()
. gen double ygrid = -2 + (_n-1) if _n <= 5
. lpbwcd y x, grid(ygrid) xeval(0) genvars(bw)
```

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